

AP 1 Precalculus 2020–2021

Unit 1: Polynomial and Rational Functions

AP Exam Weighting: 30–40% | Class Periods: 30–40

Unit Essential Questions:

- How do the algebraic structure of a polynomial or rational function reveal key features of its graph?
- When is a function model the right tool for a contextual scenario, and what assumptions must we articulate?

What's in This Packet

#	Lesson Title	AP Topic	AP Skill
1.1	Change in Tandem	1.1	2.B
1.2	Rates of Change	1.2	2.A
1.3	Rates of Change in Linear and Quadratic Functions	1.3	2.A, 2.B
1.4	Polynomial Functions and Rates of Change	1.4	2.A, 3.A
1.5	Polynomial Functions and Complex Zeros	1.5	1.B, 2.B
1.6	Polynomial Functions and End Behavior	1.6	3.A
1.7	Rational Functions and End Behavior	1.7	1.B, 3.A
1.8	Rational Functions and Zeros	1.8	1.A
1.9	Rational Functions and Vertical Asymptotes	1.9	3.A
1.10	Rational Functions and Holes	1.10	3.C
1.11	Equivalent Representations of Polynomial and Rational Expressions	1.11	1.B, 3.B
1.12	Transformations of Functions	1.12	1.C, 3.A
1.13	Function Model Selection and Assumption Articulation	1.13	2.A, 3.C
1.14	Function Model Construction and Application	1.14	1.C, 3.B
Sample Test			

AP Mathematical Practices — Unit 1 Focus

MP 1.A	Solve equations and inequalities analytically, with and without technology.
MP 1.B	Express functions, equations, or expressions in analytically equivalent forms.
MP 1.C	Construct new functions using transformations, compositions, inverses, or regressions.
MP 2.A	Identify information from graphical, numerical, analytical, and verbal representations.
MP 2.B	Construct equivalent graphical, numerical, analytical, and verbal representations.
MP 3.A	Describe the characteristics of a function with varying levels of precision.
MP 3.B	Apply numerical results in a given mathematical or applied context.

MP 3.C

Support conclusions or choices with a logical rationale or appropriate data.